

The Sovereign CTO Stack

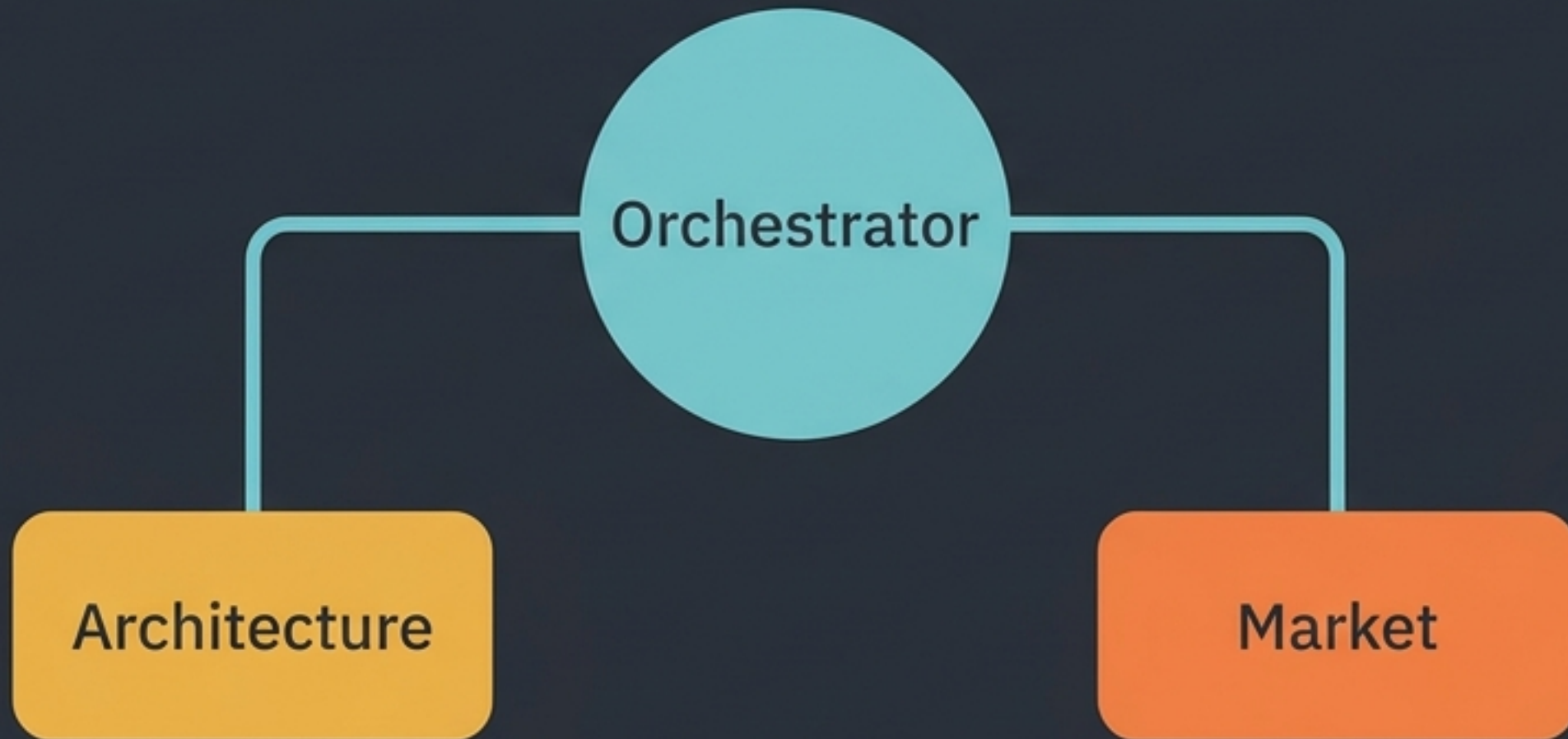
A Public-Safe, Version-Controlled Multi-Agent Engineering Factory



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  const { context } = use 'arket';

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}
```

Sovereign Sandbox



```
export Bockscodfocetn {
  const { context } = use 'arket';

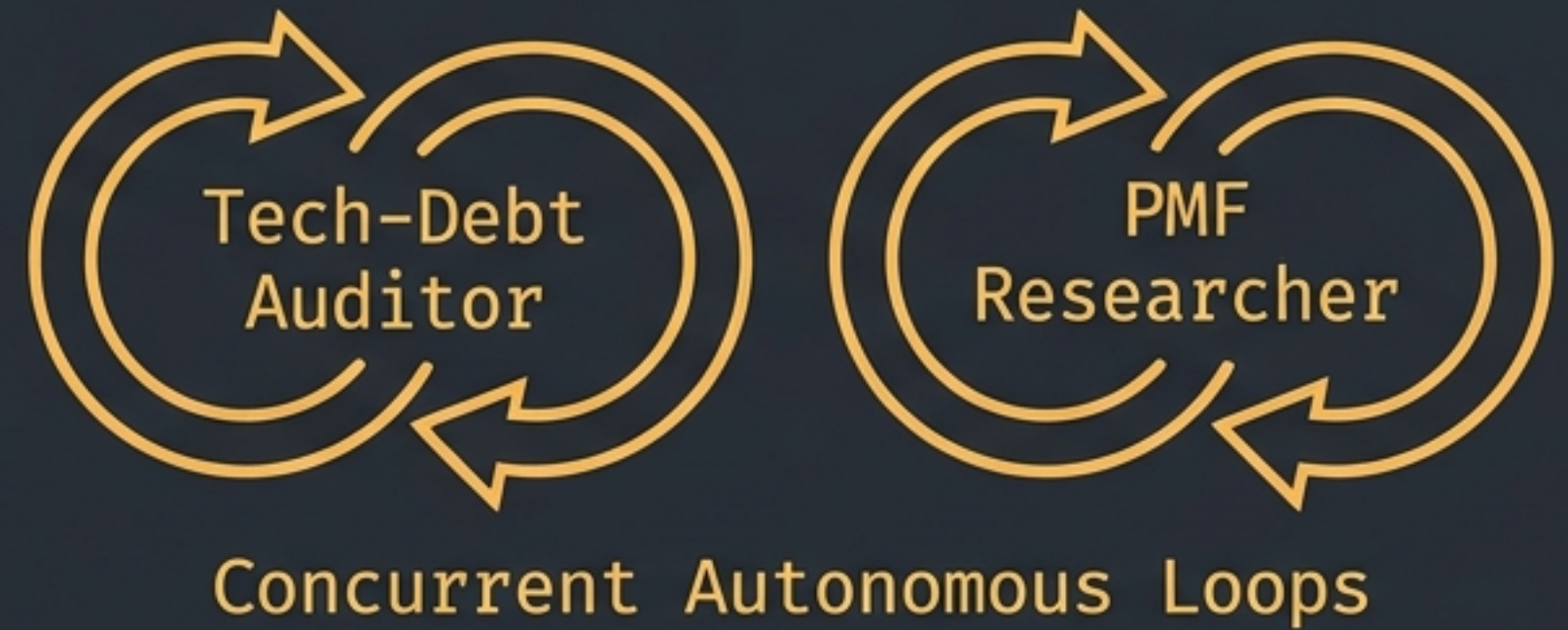
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      }
    }
  }
}
```


[■] The Brain



[■] The Specialists

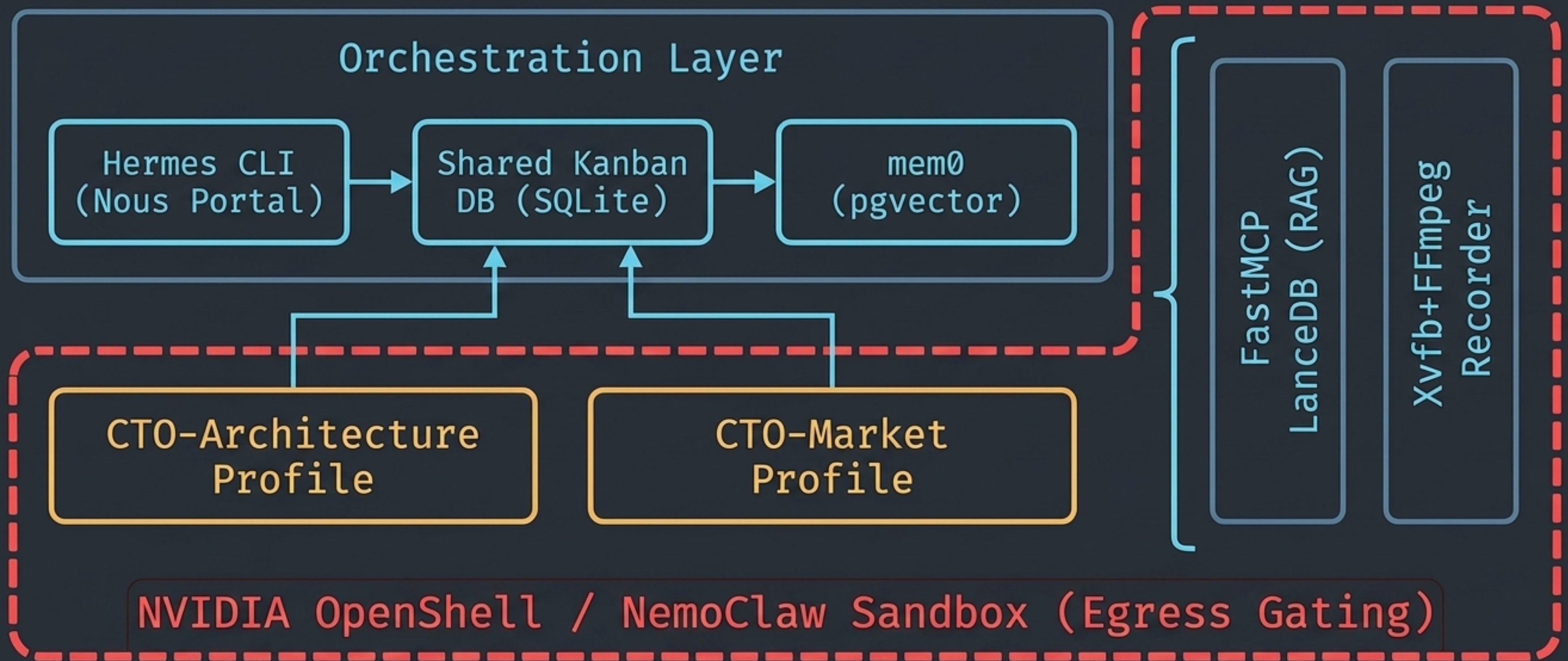


[■] The Safety



[■] The Output





Key Insight: Deterministic coordination over a single-host Kanban board rather than complex cross-host fan-outs.

The CTO RAG Brain: Grounding First



SOUL.md Non-Negotiable Rule

Agents are blocked from executing CTO functions until they issue multiple RAG queries across different dimensions and cite the exact source chunks in their final artifact.

Grounded in: `software-architecture.md`, `managing-technical-debt.md`

The Knowledge Matrix: Mapping Domains to Textbooks

The 20+ converted textbooks feeding the local Vector RAG MCP sidecar.

Tech-Debt / Architecture

Building Microservices, The Hard Parts,
Designing Data-Intensive Applications

Delivery / Flow

Accelerate, Lean Enterprise

PMF / Growth

The Lean Product Playbook, Hacking
Growth

Org Design

Team Topologies, An Elegant Puzzle

Hero Loop 1: The Tech-Debt Auditor



Signal Synthesis: Fusing Quality with Coupling

SonarQube (DETECT)

Finds 240 issues (230 smells, 7 bugs, 3 vulns).

Missing signal: Lacks cross-service coupling and blast radius context.

Graphify (KEEP)

Maps precise node degrees (frontend=7, checkout=6).

Missing signal: Blind to code quality and vulnerabilities.

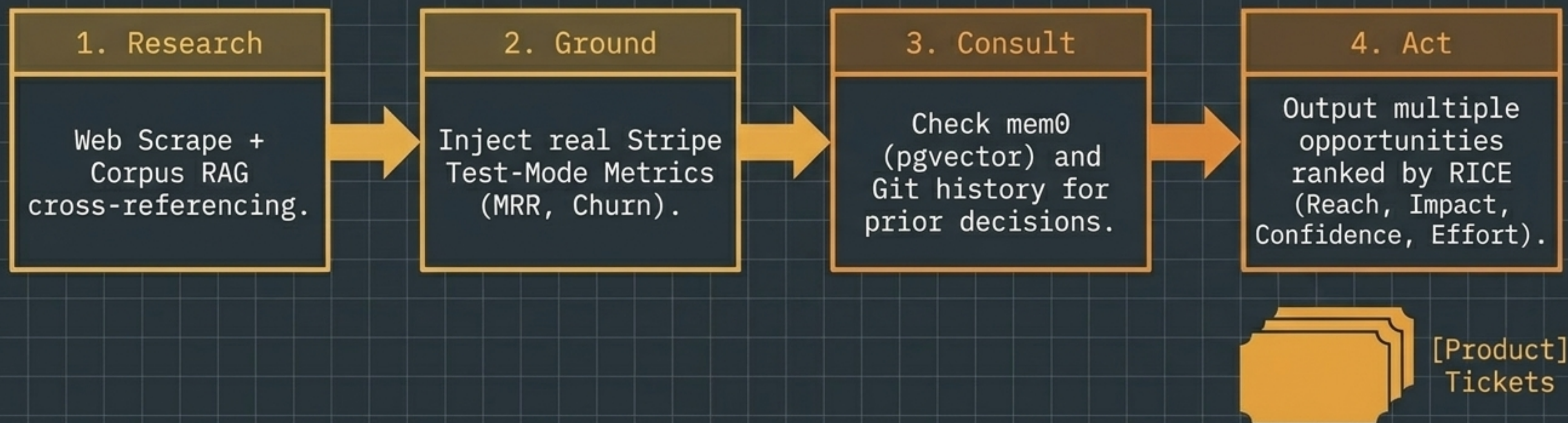
Hermes (JUDGMENT)

Fuses both signals into service-coupling.json.



Prioritizes the high-coupling, high-issue Billing Path (checkout service) due to revenue risk.

Hero Loop 2: Product-Market Fit (PMF)

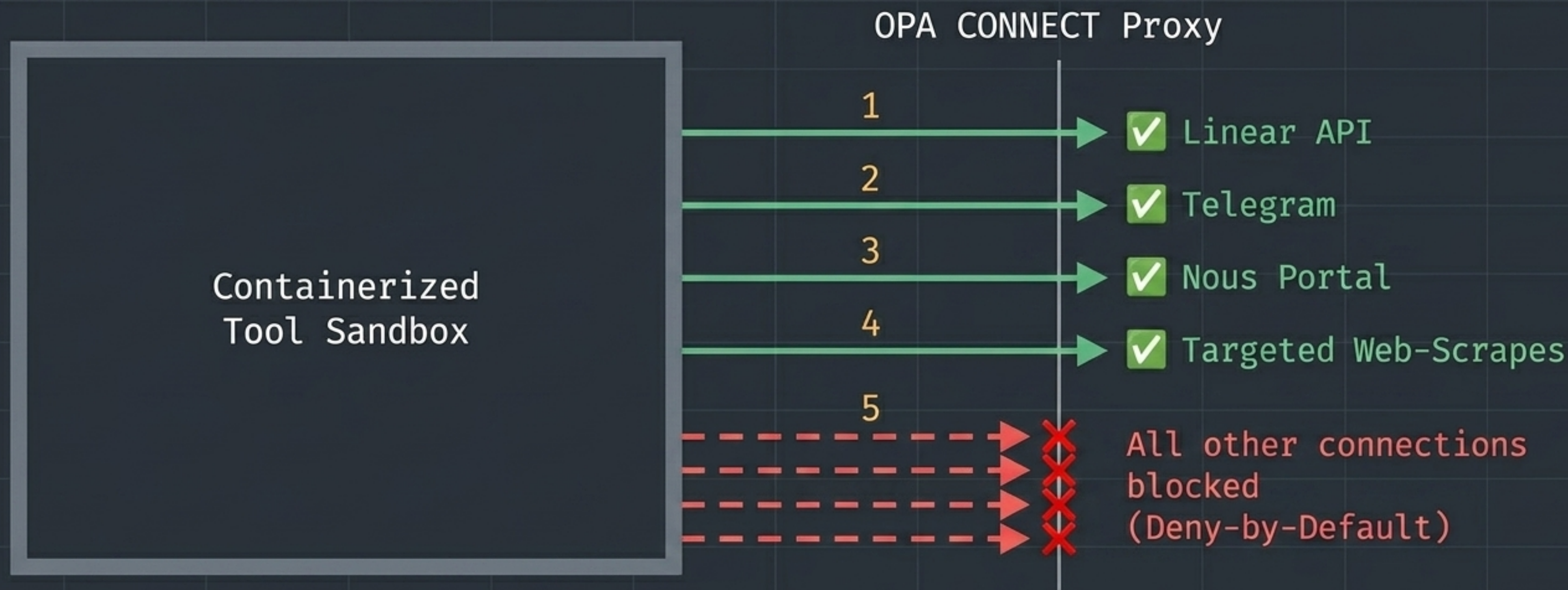


Grounding in Reality: The Data Proof

Expected (Generic AI)	Actual (Sovereign Stack)
AARRR Revenue	
Scrapes competitor pricing pages for generic assumptions.	Grounds in <code>real Stripe</code> test-mode data: <code>\$1,281/mo MRR</code> , <code>25%</code> lifetime churn.
Ideation & Memory	
Unconstrained LLMs endlessly <code>hallucinate</code> repetitive ideas.	Consults <code>mem0</code> prior-decision memory. Explicitly drops already-decided bets (e.g., ticket <code>GLO-GLO-12</code>) to avoid redundant work.

The Sovereign Safety Layer

Out-of-process enforcement via NVIDIA OpenShell and NemoClaw



Proving Sovereignty: The Negative Test

Verified via an enforced sandbox, independent of macOS kernel constraints.

Positive Path

```
> curl api.linear.app
```

```
HTTP 200 OK
```

The Negative Test

```
> curl example.com
```

```
(56) CONNECT tunnel failed,  
response 403
```

The Insight: The load-bearing assertion is the denial.
It proves the sandbox actively restricts rogue activity.

Observable Execution: The Capture Pipeline

Pixel-perfect .mp4 capture verified by strict CI checks.

Live Agent Terminal

```
> agent.tool_executor: tool mcp_cto_knowledge...  
| executed with args: {'query': 'Linear ticket  
creation flow'} | timestamp: 2023-10-27T10:45:22Z
```

[INFO] Action detected: Create Linear ticket

[DEBUG] Navigating to Linear API

[DEBUG] Navigating to Linear API

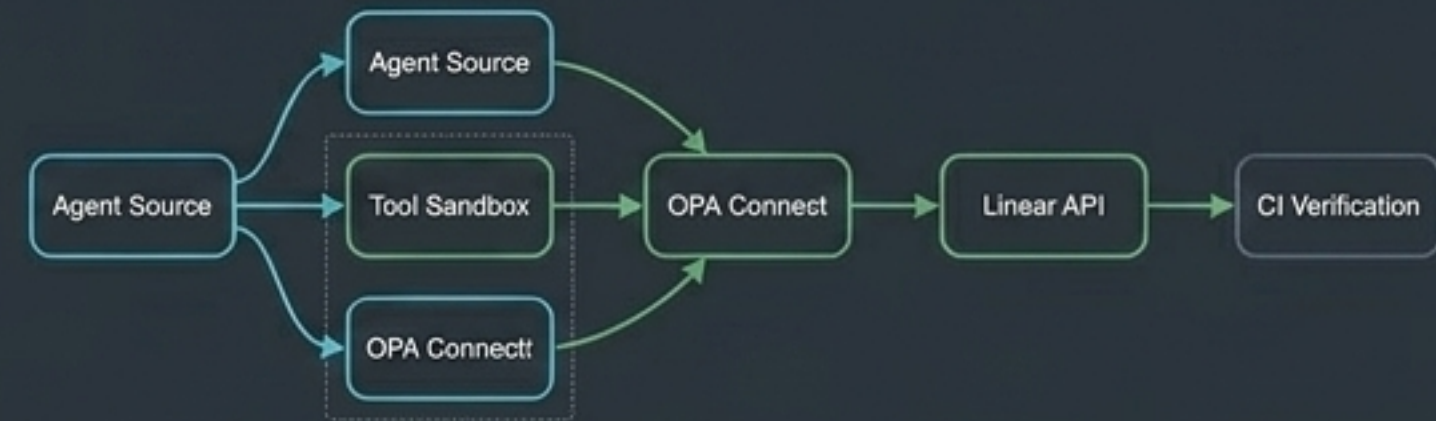
[DEBUG] Navigating to Loan Service

HTTP Request: POST /graphql (200 OK)

HTTP Request: POST /graphql (200 OK)

HTTP Request: POST /graphql (200 OK)

Rendered HTML View



Generated Linear Ticket [Brownfield]

Title

Automate workflow

Description

Based on knowledge here early to the workacholide gandomcreating this zwente enroty and oversodate the linear ticket ridzer workflow.

Status

Triage

Priority

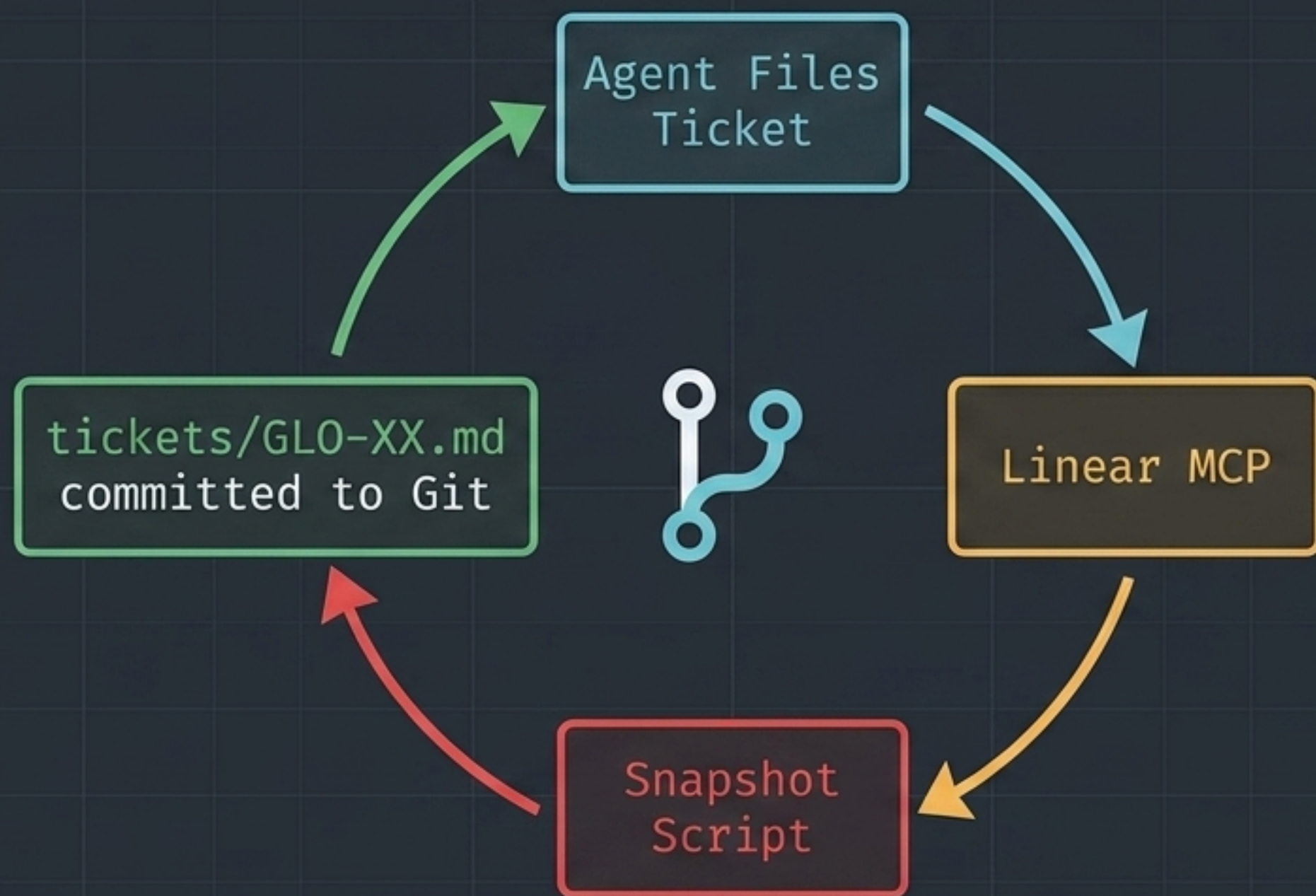
Medium

Assignee

unassigned

Create

Git as the Authoritative Decision Record



- mem0 AI memory is utilized strictly as a recall convenience.
- Git is the durable, auditable source of truth. The repo remains clean-cloneable, public-safe, and gated by fail-closed gitleaks hooks.

The Full-Build Epic: Executing the Backlog

GL0-13 Achievements (Phases 0-5 + P1-P4)

- [x] Real Tool Calls (P0): Authentic log streaming.
- [x] OpenShell Egress (P1): Proven negative-test sandbox denial.
- [x] Stripe Grounding (P2): Real MRR & Churn data integration.
- [x]
- [x] SonarQube Fusion (P3): Quality metrics layered onto Graphify coupling.
- [x] Ranked PMF + mem0 (P4): RICE-ranked opportunities aware of prior bets.
- [x] Showcase Montage: Hybrid pixel-capture video assembly.

The Future State (GLO-14)

The Sovereign CTO Stack is an executing, version-controlled roadmap.

